

ELI5: ai and water usage

Posted by u/chasincloudz • 0 points • 7 comments

how does ai, something that creates 'art' take away from natural resources, what is the correlation

Comments

u/dale_glass • 1 points • 2024-11-25 09:36 UTC

A thing that's confusing in this area is that "AI" has a very wide scope.

Artwork generation is "simple". It's doable in home conditions for inference (creating images) and training on small scales. It's very comparable to gaming usage-wise. It can certainly increase your power bill if you try hard enough but is not that special, you already could heat your room by gaming.

LLMs (large language models) are the big datacenter and water usage kinds. Those are the ones that talk to you, like ChatGPT. They're far bigger and take way fancier hardware than image generation. That's where you see huge costs, big power bills, and water usage for cooling it all.

u/EmergencyCucumber905 • 1 points • 2024-11-24 22:27 UTC

The chips in datacenters generate heat. Some datacenters use water to cool the chips. Cold water is pumped through the system, heated by the chips then pumped into a reservoir for cooling. Most of the time this system is a closed loop, so water isn't really being wasted.

u/DarkAlman • 2 points • 2024-11-24 20:36 UTC

AI tools require an immense amount of processing power, usually in the form of GPUs (video card processors).

This requires a lot of space, consumes a lot of electricity, and produces a lot of waste heat that needs to be gotten rid of with some kind of cooling.

There is also a lot of overlap between AI research and Crypto currencies as they use similar hardware.

Both could be considered an ecological disaster consider how much electricity, precious metals, and resources they consume.

u/MaverickGuardian • 1 points • 2024-11-24 20:31 UTC

Data centers should exist mostly in cool climate cities so that excess energy could be used for heating near by buildings. But yeah, then the latency increases.

u/saul_soprano • 3 points • 2024-11-24 20:16 UTC

AI requires a lot of electricity. Common energy sources such as coal, fossil fuels, and nuclear use water whether it's for cooling or heat extraction.

u/Nerdsamwich • 1 points • 2025-02-12 07:30 UTC

But those commonly use a closed-loop system, don't they? I thought power plants recycled their cooling water.

u/Mono_Clear • -1 points • 2024-11-24 20:14 UTC

Power, if you're dealing with a zero-sum gain than you can equate everything to the usage of power.

If you're quantifying productivity in kilowatts than any kilowatt spent doing something you think is a waste of time is a kilowatt taking away from something you think is useful.

u/ryschwith • 5 points • 2024-11-24 20:14 UTC

AI requires a very large amount of processing power, which consumes a lot of electricity and generates a lot of heat. Water is frequently used for cooling in large data centers (and sometimes in power generation).

u/Nerdsamwich • 1 points • 2025-02-12 07:28 UTC

Okay, but we mastered closed-loop cooling systems over a century ago. Why would they not be used in a server farm?

u/ryschwith • 1 points • 2025-02-12 15:07 UTC

I'm by no means an HVAC specialist, but if I had to guess I'd say that closed-loop cooling systems probably aren't efficient enough to support the massive amounts of heat generated by a data center or server farm.

u/Nerdsamwich • 1 points • 2025-02-12 17:21 UTC

It's good enough for a nuclear reactor.